DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

EVALUATION SCHEME & SYLLABUS FOR

B. TECH. III YEAR

AERONAUTICAL ENGINEERING

ON

CHOICE BASED CREDIT SYSTEM (CBCS)

[Effective from the Session: 2018-19]

STUDY AND EVALUATION SCHEME

B-Tech. Aeronautical Engineering

YEAR: 3rd / SEMESTER-V

S. No.	Subject Code	Subject Name	Teaching Department	L-T-P	Theory / Lab Marks	SESSIONAL		Total	Credit
						Test	Assignment / Attendance		
1	RAS501	Managerial Economics	Applied Science	3-0-0	70	20	10	100	3
2	RAS502/ RUC501	Sociology /Cyber Security	Applied Science	3-0-0	70	20	10	100	3
3	RME502A	Heat & Mass Transfer	ME	3-0-0	70	20	10	100	3
4	RAE501	Aircraft Instruments & Systems	AE	3-1-0	70	20	10	100	4
5	RAE502	Design of Aircraft Elements	AE	3-0-0	70	20	10	100	3
6	RAE051 -054	Deptt. Elective Course-1	AE	3-1-0	70	20	10	100	4
7	RME552	Heat & Mass Transfer Lab	ME	0-0-2	50		50	100	1
8	RAE551	Aircraft Instruments Lab	AE	0-0-2	50		50	100	1
9	RAE552	Introduction to Aeronautical Engg. Lab	AE	0-0-2	50		50	100	1
10	RAE559	Seminar – I		0-0-2			100	100	1
TOTAL								**1000**	**24**

Department Electives-1 Courses:

1. **RAE051- Introduction to Aerospace Propulsion**
2. **RAE052- Space flight Mechanics**
3. **RAE053–Air Navigation**
4. **RAE054- Unconventional Manufacturing Process.**

STUDY AND EVALUATION SCHEME

B-Tech. Aeronautical Engineering

YEAR: 3rd / SEMESTER-VI

S. No.	Subject Code	Subject Name	Teaching Department	L-T-P	Theory / Lab Marks	SESSIONAL		Total	Credit
						Test	Assignment / Attendance		
1	RAS601	Industrial Management	Applied Science	3-0-0	70	20	10	100	3
2	RAS602/ RUC601	Sociology /Cyber Security	Applied Science	3-0-0	70	20	10	100	3
3	RME602A	Theory of Machines	ME	3-0-0	70	20	10	100	3
4	RAE601	Aircraft Structure	AE	3-1-0	70	20	10	100	4
5	RAE602	Aerodynamics-I	AE	3-0-0	70	20	10	100	3
6	RAE061 -064	Deptt. Elective Course-2	AE	3-1-0	70	20	10	100	4
7	RME652	Theory of Machines Lab	ME	0-0-2	50		50	100	1
8	RAE651	Aircraft Structure Lab	AE	0-0-2	50		50	100	1
9	RAE652	Aircraft Propulsion Lab	AE	0-0-2	50		50	100	1
10	RAE653	Design of Aircraft Elements& Simulation Lab	AE	0-0-2	50		50	100	1
TOTAL								**1000**	**24**

Department Electives-2 Courses:

1. **RAE061- Jet Aircraft Propulsion**
2. **RAE062- Composite Material & Structure**
3. **RAE063- Aircraft material & Non-Destructive Testing**
4. **RAE064- Aircraft Evaluation**

RME----: HEAT & MASS TRANSFER

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UNIT-1

Introduction to Heat Transfer:

Thermodynamics and Heat Transfer. Modes of Heat Transfer: Conduction, convection and radiation. Effect of temperature on thermal conductivity of materials; Introduction to combined heat transfer mechanism.

2

Conduction:

General differential heat conduction equation in the rectangular, cylindrical and spherical coordinate systems.Initial and boundary conditions.

3

Steady State one-dimensional Heat conduction:

Simple and Composite Systems in rectangular, cylindrical and spherical coordinates with and without energy generation; Concept of thermal resistance.Analogy between heat and electricity flow; Thermal contact resistance and over all heat transfer coefficient; Critical radius of insulation.

3

UNIT-2

Fins:

Heat transfer from extended surfaces, Fins of uniform cross-sectional area; Errors of measurement of temperature in thermometer wells.

3

Transient Conduction:

Transient heat conduction; Lumped capacitance method; Time constant; Unsteady state heat conduction in one dimension only, Heisler charts.

UNIT-3

Forced Convection:

Basic concepts; Hydrodynamic boundary layer; Thermal boundary layer; Analogy between momentum and heat transfer in turbulent flow over a flat surface; Mixed boundary layer; Flow over a flat plate; Flow across a single cylinder and a sphere; Flow inside ducts; Empirical heat transfer relations;

5

Natural Convection:

Physical mechanism of natural convection; Buoyant force; Empirical heat transfer relations for natural convection over vertical planes and cylinders, horizontal plates and cylinders, and sphere, Combined free and forced convection.

5

UNIT-4

Thermal Radiation:

Basic radiation concepts; Radiation properties of surfaces; Black body radiation Planck's law, Wein's displacement law, Stefan Boltzmann law, Kirchoff's law; ; Gray body; Shape factor;

Black-body radiation; Radiation exchange between diffuse non black bodies in an enclosure; Radiation shields; Radiation combined with conduction and convection; Absorption and emission in gaseous medium; Solar radiation; Greenhouse effect.

8

UNIT-5

Heat Exchanger:

Types of heat exchangers; Fouling factors; Overall heat transfer coefficient; Logarithmic mean temperature difference (LMTD) method; Effectiveness-NTU method; Heat transfer enhancement (augmentation) methods.

3

Condensation and Boiling:

Introduction to condensation phenomena; Heat transfer relations for laminar film condensation on vertical surfaces and on outside & inside of a horizontal tube; Effect of non-condensable gases; Dropwise condensation; Heat pipes; Boiling modes, pool boiling; Two phase flow and heat transfer.

3

Introduction to Mass Transfer:

Introduction; Fick's law of diffusion; Steady state equimolar counter diffusion; Steady state diffusion though a stagnant gas film. Combined heat and mass transfer Micro scale and nano scale heat & mass transfer.

2

Books and References:

1. Fundamental of Heat and mass transfer, by Incroperra & De witt John wiley sons.
2. Heat Transfer by J.P. Holman, McGraw-Hill.
3. Heat and Mass Transfer by Rudramoorthy and Mayilsamy, Pearson Education.
4. Heat Transfer by Ghoshdastidar, Oxford University Press .
5. A text book on Heat Transfer, by Sukhatme, University Press.
6. Heat Transfer by Venkateshan, Ane Books Pvt Ltd.
7. Schaum's outline of Heat Transfer by Pitts & Sisson McGraw-Hill
8. .Heat and Mass Transfer by R Yadav, Central Publishing House

RAE 501: AIRCRAFT INSTRUMENTS& SYSTEMS

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UNIT-1 **8**

Electronic Instrument System: Typical system arrangement and cockpit layout of Electronics instrument systems. Numbering System: binary, octal and hexadecimal: Demonstration of conversion between the decimal And binary, octal and hexadecimal systems and vice versa. Logic Circuits: Identification of common logic gate symbols and equivalent circuits.
Application used for aircraft systems.

UNIT-2 **8**

Data Conversion: Analogue Data, Digital Data: Operation and application of analogue to digital, and digital to analogue converters, inputs and outputs, limitations of various types Data Buses: Operation of data buses in aircraft systems, including knowledge of ARINC and other specifications. schematic diagrams. Interpretation of logic diagrams.

UNIT-3 **9**

Microprocessors: Functions performed and overall operation of a microprocessor. Basic operation of each of the following microprocessor elements: control and processing unit, clock, register, arithmetic logic unit. Integrated Circuits, Operation and use of encoders and decoders; Function of encoder types; Uses of medium, large and very large scale integration. Multiplexing. Operation, application and identification in logic diagrams of multiplexers and DE multiplexers.

UNIT-4 **9**

Fiber Optics: Advantages and disadvantages of fiber optic data transmission over electrical wire propagation: Fiber optic data bus, Fiber optic related terms, Termination, Couplers, control terminals, remote terminals. Application of fiber optics in aircraft systems.

Electronic Displays: Principles of operation of common types of displays used in modern aircraft, including Cathode Ray Tubes, Light Emitting Diodes and Liquid Crystal Display. Electrostatic Sensitive Devices. Special handling of components sensitive to electrostatic discharges. Awareness of risks and possible damage, component and personnel anti-state protection devices. Software Management Control.

UNIT-5 **10**

Electromagnetic Environment: Influence of the following phenomena on maintenance practices for electronic system: EMC-Electromagnetic Compatibility EMI Electromagnetic Interference HIRF-High Intensity, Radiated Field. Lightning, lightning protection. Typical Electronic/Digital Aircraft Systems. General arrangement of typical electronics.

Digital aircraft system and associated BTTE(Built in Test Equipment): Testing such as: ACARS-ARINC Communication and Addressing and Reporting System, ECAM- Electronics.
Centralized Aircraft Monitoring. EFIS- Electronics Flight Instrument System, EICAS- Engine Indication and Crew Alerting System, FBW-Fly by Wire, FMS- Flight Management System, GPS- Global Positioning System & IRS-Inertial Reference System.

Books and References:

1. Pallet, E.H.J, "Aircraft Instruments & Principles", Pitman & Co 1993.
2. Casamassa, J.V., and Bent R., Jet Aircraft Power Systems, McGraw Hill Book Co., New York,1975
3. Bent, R.D. Mickinely., Aircraft Maintenance and Repair, 2nd Edition – McGraw Hill Inc ., NowYork, 1978.

RAE-502: DESIGN OF AIRCRAFT ELEMENTS

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UNIT-1 8

Introduction: Definition, Design requirements of machine elements, Design procedure, Standards in design, Selection of preferred sizes, Indian Standards designation of carbon & alloy steels, Selection of materials for static and fatigue loads.

Design for Static and Fluctuating Loads: Modes of failure, Factor of safety, Principal stresses, Stresses due to bending and torsion, Theory of failure. Cyclic stresses, Fatigue and endurance limit, Stress concentration factor,Stress concentration factor for various machine parts, Notch sensitivity, Design for finite and infinite life, Soderberg & Goodman & Gerber criteria.

UNIT-29

Riveted Joints: Riveting methods, materials, Types of rivet heads, Types of riveted joints, Caulking and Fullering, Failure of riveted joint, Efficiency of riveted joint, Design of boiler joints, Eccentric loaded riveted joint.

Keys and Shafts: Cause of failure in shafts, Materials for shaft, Stresses in shafts, Design of shafts subjected to twisting moment, bending moment and combined twisting and bending moments, Shafts subjected to fatigue loads, Design for rigidity.Types of keys, splines, Selection of square & flat keys, Strength of sunk key.

UNIT-39

Spur Gear: Principle of transmission and conjugate action**.**Tooth forms, System of gear teeth, contact ratio, Standard proportions of gear systems, Interference in involute gears, Backlash, Selection of gear materials, Gear manufacturing methods, Design considerations, Beam strength of gear tooth, Dynamic tooth load, Wear strength of gear tooth, Failure of gear tooth, Design of spur gears, AGMA and Indian standards.

Helical Gears: Terminology, Proportions for helical gears, Forces components on a tooth of helical gear, Virtual number of teeth, Beam strength& wear strength of helical gears, Dynamic load on helical gears, Design of helical gears.

UNIT-4 8

Bevel and Worm Gears: Terminology of bevel gears, Force analysis, Virtual number of teeth, Beam strength and wear strength of bevel gears, Effective load of gear tooth, Design of a bevel gear system.Types of worms, Terminology, Gear tooth proportions, Efficiency of worm gears, Heat dissipation in worm gearing, Strength and wear tooth load for worm gears, Design of worm gearing system.

UNIT-510

Sliding Contact Bearing: Types, Selection of bearing, Plain journal bearing, Hydrodynamic lubrication, Properties and materials, Lubricants and lubrication, Hydrodynamic journal bearing, Heat generation, Design of journal bearing, Thrust bearing-pivot and collar bearing, Hydrodynamic thrust bearing.

Rolling Contact Bearing

Types of ball bearing, Thrust ball bearing, Types of roller bearing, Selection of radial ball bearing, bearing life, Selection of roller bearings, Dynamic equivalent load for roller contact bearing under constant and variable loading, Reliability of Bearing, Selection of rolling contact bearing, Lubrication of ball and roller bearing. Advantages and disadvantages.

Note: Design data book is allowed in the examination.

Books and References:

1. Design of Machine Elements-V.B. Bhandari, Tata McGraw Hill Co.
2. Machine Design-Sharma and Agrawal, S.K. Kataria& Sons.
3. Machine Design, U C Jindal, Pearson Education.
4. Machine Design, Sadhu SIngh, Khanna Book Publishing Co.
5. Design of Machine Elements, Sharma and Purohit, PHI.
6. Design of Machine Eesign-M.F. Spott, Pearson Eductaion
7. Machine Design-Maleev and Hartman, CBS Publishers.
8. Machine Design Data Book, Sadhu Singh, Khanna Book Publishing Co.
9. Mechanical Engineering Design, 9e – Joseph E. Shigely, McGraw Hill Education
10. Elements of Machine Component Design, Juvinal&Marshek, John Wiley & Sons

DEPARTMENT ELECTIVE-1

RAE-011:INTRODUCTION TO AEROSPACE PROPULSION

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UNIT-1 **8**

Introduction: Course Intro & Historical development of flightsEarly development of aircraft propulsive devicesDevelopment of Jet propulsion for aircraft Introduction to thermodynamics, Scope and method, Basic concepts: system, surroundings, property, intensive and extensive, state, equilibrium and state postulate, process, path and cycleQuasi-static processes, zeroth law of thermodynamics and temperature, concept of energy and its various forms, internal energy, enthalpySpecific heats at constant pressure and volume Work and heat transfers.

UNIT-2 **8**

First law of thermodynamics for closed systems.Processes.Second law of thermodynamics, heat engines, refrigerators and heat pumps, Kelvin-Planck and Clausius statement of second law of thermodynamics.Reversible and irreversible processes, concept of entropy.Increase of entropy principle, third law of thermodynamics, absolute entropy, perpetual motion machines. Carnot cycle, Carnot principle, thermodynamic temperature scale.Energy, availability and second law efficiency

UNIT-39

Cycles Analysis: Gas and vapor power cycles, Otto cycle, Diesel cycle, Dual cycle.Rankin cycle, Brayton cycle, Sterling and Ericsson cycles. Thermodynamic property relations, Jacobean and Legendre transformations, Maxwell's equations.Properties of gas and vapor mixtures. One-dimensional compressible flows, isentropic flows.Flows with friction and heat transfer, normal and oblique shocks

UNIT-49
IC Enginesand Jet Engines for aircraft workingand application.Performance parameters of IC engines.
Supercharging of aircraft IC engines.Propeller fundamentals, Propeller aerodynamic theories – I.Propeller aerodynamic theories –II.
Ideal cycles for Jet engines.Ideal cycles for variants of jet engines.Fundamentals of Ramjets and Pulsejets.Fundamentals of Rocket engines.Fundamentals of Missile engines.Various space vehicles and their engines.

UNIT- 510
Combustion in S.I. Engine: Normal combustion in S.I. engine, Auto ignition of end charge,Knock and detonation, Pre-ignition and post-ignition. Pressure-specific volume (p.v.) and pressure- time(p.t.) diagrams for normal combustion.P.T. and rate diagrams with detonation. Effect of engine operating variables on knock, Knock rating of S.I. Fuels, Antiknock agents,

Knock and S.I. engine performance.
Ignition: Ignition timing and performance. Effect of ignition timing on output. Factors effectingspark advance.Ignition timing and cylinder temperature. Location of spark plug, battery ignition system. Magneto ignition Aircraft engine plugs. Altitude effects on ignition apparatus.

Books and References:

1. Obert, E.F., Internal Combustion Enginers, Edn. 3, International Text Book Company, 1969.
2. Judge, AW, Modern Petrol Engine, Edn. 2, Chapman and Hall Ltd., 1955.
3. Regowski, AR. Elements of Internal-Combustion Engines, McGraw Hill Book Co Inc. 1953.
4. Lichty L.C., Internal Combustion Engines, Edn. 6, McGraw Hill Book Co., Inc. 1951.
5. Pye, D.R., The Internal Combustion Engine, Vol; I and Vol. II, Oxford University Press,1953.
6. Morley, AW, Aircraft Propulsion - Theory and Performance, Longmans, Green and Co., 1953.

RAE- 012 SPACE FLIGHT MECHANICS

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UNIT- 1 **8**

BASIC CONCEPTS- Peculiarities of space environment and its description– effect of space environment on materials of spacecraft structure and astronauts- manned space missions – effect on satellite life time. The solar system – reference frames and coordinate systems – terminology related to the celestial sphere and its associated concepts – Kepler's laws of planetary motion and proof of the laws –Newton's universal law of gravitation - the many body problem The celestial Sphere.

UNIT- 2 **8**

THE GENERAL N-BODY PROBLEM-The Ecliptic Motion of Vernal Equinox – Sidereal Time – Solar Time –Standard Time – The Earth's Atmosphere. Study the basic concepts of orbital Mechanics with particular emphasis on interplanetary trajectories. The many body Problem – Lagrange, Jacobian identity The Circular Restricted Three Body Problem – Libration Points – Relative Motion in the N-body Problem – Two – Body Problem – Satellite Orbits – Relations Between Position and Time – Orbital Elements.

UNIT- 3 **8**

SATELLITE INJECTION AND SATELLITE ORBIT PERTURBATIONS: General Aspects of satellite Injections, Satellite Orbit Transfer, Various Cases – Orbit Deviations Due to Injection – Errors – Special and General Perturbations – Cowell's Method – Encke's Method - Method of vibrations of Orbital Elements- General Perturbations Approach.

UNIT –4 **8**

INTERPLANETARY TRAJECTORIES-Two Dimensional Interplanetary Trajectories – Fast Interplanetary Trajectories – Three Dimensional Interplanetary Trajectories – 3-Dimensional Interplanetary Trajectories – Launch if Interplanetary Spacecraft – Trajectory estimation about the Target Planet. Concept of the sphere of influence. Lamberts theorem.

UNIT- 5 **8**

BALLISTIC MISSILE TRAJECTORIES AND MATERIALS: The Boost Phase – The Ballistic Phase – Trajectory Geometry – Optimal Flights – Time of Flight – Re-entry Phase – The Position of the Impact Point – Influence Coefficients. Space Environment - Peculiarities - Effect of Space Environment, the Selection of Spacecraft Material.

Books and Reference:

1. Cornelisse, J.W. "Rocket Propulsion and Space Dynamic", W.H. Freeman & Co., 1984.
2. Sutton, G.P., "Rocket Propulsion Elements", John Wiley, 1993.

3. Van de Kamp, P., “Elements of Astro-mechanics”, Pitman, 1979.
4. Parker E.R., “Material for Missiles and Spacecraft”, McGraw – Hill Book Co., Inc., 1982.

RAE- 013: AIR NAVIGATION

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UNIT-1 **8**

The Problems of Air Navigation: The Aircraft, Aids of Navigation VOR, ADF, ILS, GCA, TACAN, Doppler and basics of inertial navigation: their limitations and uses, Weather, Air Traffic Control Communication.

Chart Projection for Air Navigation: The Round Earth on a Flat Chart, Properties obtainable in Projection, Distance on a Sphere, Direction on a Sphere, The Lambert Projection, The Mercator Projection, The Gnomonic Projection, The Stereographic Projection, and Projections for Weather Charts. Calculation on Rhomb Line Tracks and distances.

UNIT-2 **8**

Magnetism: Review of the Elementary laws of magnetism, Terrestrial magnetism, Horizontaland vertical components of earth's magnetic field and their variation with latitude Isogonic and agonic lines. Isoclinic lines. Aircraft Magnetism, Resolution into P.Q. and R components, coefficients and deviation associated with them, compass course deviation.

Instruments:- Units of measurement of distances and height, The function of navigationalInstruments, The Speed Indicator, The Rate of Climb indicator, The Altimeter, The Magnetic Compass, The Turn and Bank Indicator, The Directional Gyro, The Artificial Horizon, Radio,
Radar Altimeter, Mach meter& Fluxgate Compass.

UNIT-3 **8**

Chart Reading: Distinctive Properties of Charts, The importance of chart reading,Topographic Information Aeronautical Data, Legend and written Notes, The practice of Chart Reading.

Dead Reckoning: The place of Pilotage, Advantage of Dead Reckoning, Basic Problems inDead Reckoning.

Special Problems & Dead Reckoning: Climb and Descent, off course corrections, DoubleDrift. Radius of Action, Cruise Control, Alternate Airport Problem, Interception, Tracking.

Air Navigation Computers: Function and Usefulness,the Slide' Rule Side the Wind TriangleSide.

UNIT-4 **8**

Radio Navigation: Principles of radio transmission and reception, properties ofelectromagnetic waves, classification of frequency bands, elementary knowledge of Radar.An elementary knowledge of principles of the following radio and radar aids and systems: Airborne D/F The manual loop and automatic radio compass including methods of calibration Ground D/F M.F., H.F. and V.H.F. Systems.Radio/Radar track guides approach and landing aids and systems including V.O.R., N.D.B. and I.L.S.Plotting Radio Bearings on Mercator Charts.

UNIT-5 **8**

Celestial Navigation: Elements of Astronomy, the universe, Solar system, movements of earth,moon and planets, earth's orbit' Kepler laws, declination, altitude, azimuth etc. Practical Value, Accuracy, Simplicity, Basic principles, The Line of position, Celestial Coordinates,

Determining the Greenwich Hour Angle (GHA) Determining the Local Hour Angle (LHA). Aline of Position from Polaris. A Line of Position from and Unidentified Star, Radio Time Signals, Identification of Stars, Star Names, Brightness of Star, the planets. Motion of the Stars and Planets. The Practice of Celestial Navigation, Astro-Navigation instruments.

The Practice of Navigation: Details of Navigation. Preparation of Charts for use in FlightPilotage Contact Instrument Flying.

Books and References:

1. C.W. Martin, Air Navigation.
2. Weeins, Air Navigation.
3. D.C.T. Benett, TheComplete Air Navigation.
4. T.C. Lyon, Practical Air Navigation.
5. RAT Manual of Air Navigation, A.P. 1234Vols. A.B.D. & E.
6. Martin, Air Navigation
7. Brown, Air Navigation, Chaman and Hall.
8. Brook, Williams and Branch, Air Navigation, Theory and Practice.
9. J. Clough Smith, Applied D.R. Navigation and Flight Planning.

RAE – 014: UNCONVENTIONAL MANUFACTURING PROCESSES

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UNIT-18
Introduction: Limitations of conventional manufacturing process, need of unconventionalmanufacturing processes & its classification and its future possibilities.
Electronic-device Manufacturing: Brief description of Diffusion and Photo - **Lithography process** for electronic-device manufacturing.

UNIT-2 **8**
Unconventional Machining Process: Principle and working and applications of unconventionalmachining process such as **Electro-Discharge machining, Electro-chemical machining,ultrasonic machining,** Abrasive jet machining etc.

UNIT-3 **8**
Unconventional Machining Process (continued): Principle and working and application ofunconventional machining processes such as Laser beam machining, Electron beam machining, Ultrasonic machining etc. (these can also be used for welding).

UNIT-4 **8**
Unconventional welding processes: Explosive welding, Cladding etc. Under water welding.Metalizing, Plasma are welding/cutting, gas welding / cutting & Resistance Arce welding etc.

UNIT-5 **8**
Unconventional Forming process: Principle, working and applications of High energy formingprocess such as Explosive Forming, Electromagnetic forming, Electro-Discharge forming, water hammer forming, explosive compaction etc.

Books and References:

1. Modern Machining Processes - P.C. Pandey
2. Unconventional Machining - V.K. Jain.

RME- ----: HEAT & MASS TRANSFER – LAB

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Minimum 8 out of following such Experiments (or such Experiments)

1. Conduction – Experiment on Composite plane wall
2. Conduction – Experiment on Composite cylinder wall
3 Conduction - Experiment on critical insulation thickness
4. Conduction – Experiment on Thermal Contact Resistance
5. Convection - Pool Boiling experiment
6. Convection - Experiment on heat transfer from tube-(natural convection).
7. Convection - Heat Pipe experiment.
8. Convection - Heat transfer through fin-(natural convection).
9. Convection - Heat transfer through tube/fin-(forced convection).
10 Convection - Determination of thermal conductivity of fluid
11 Experiment on Stefan's Law, on radiation determination of emissivity, etc.
12 Experiment on solar collector, etc.
13. Heat exchanger - Parallel flow experiment
14. Heat exchanger - Counter flow experiment

RAE- 551: AIRCRAFT INSTRUMENTS
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Minimum 10 out of following such Experiments (or such Experiments)

1. Safety precaution associated with radio equipment hazards: high voltage, RF emission and microwave emissions, Electrostatic discharge etc.
2. Wiring and cabling demonstration and practice in radio circuits.
3. Multimeter, Megger and bonding testers: demonstration and practice.
4. Soft and hard soldering practices.
5. Identification and inspection of antenna: external wire aerials, blade, rod and rail aerials:
6. D/F loops and suppressed aerials viewing on A/C and inspection for physical condition. Aerial masts, static discharger's etc. inspection and servicing.
7. To demonstrate the measurements TRF receiver, intermediate frequency and amplifier Frequency convertor.
8. Super heterodyne alignment Buffer amplifier and RF amplifier.
9. To demonstrate Modulation and demodulation.
10. To study of Interference (filtering and shielding) Troubleshooting Practices.
11. To study the working of Bourdon Pressure Gauge and to check the calibration of the gauge in a dead-weight pressure gauge calibration set up.
12. To measure static/dynamic pressure of fluid in pipe/tube using pressure transducer/pressure cell.
13. Vibration measurement by Dual Trace Digital storage Oscilloscope.
14. To study of the transmission losses by a given transmission line by applying capacitive /inductive load.

RAE-552: INTRODUCTION TO AERONAUTICAL ENGG. LAB

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Minimum 10 out of following such Experiments (or such Experiments)

1. To demonstrate the longitudinal, lateral and directional stability.
2. To demonstrate the effect of roll control: ailerons and spoilers:
3. To demonstrate the effect of pitch control: elevators and horizontal stabilators.
4. To demonstrate the effect of yaw control, rudder limiters:
5. To demonstrate the effect of pitch control variable incidence stabilizers and canards.
6. To demonstrate the effect of high lift devices: slots, slats , flaps
7. To demonstrate the effect of drag inducing devices: spoilers, lift dumpers, speed brakes:
8. To demonstrate the effect operation and effect of trim tabs, servo tabs, control surface bias.
9. To demonstrate the effect operation with stall protection system
10. To demonstrate the effect with aircraft major components on aircraft and to identify their location.
11. To demonstrate the effect with aircraft structure fuselage station, wing station number
12. To demonstrate the effect operation understand and test electrical bonding on aircraft
13. To demonstrate the effect operation with manual system operation
14. To demonstrate the effect operation with hydraulic actuator

RME---- THEORY OF MACHINES **L T P**

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UNIT-1

8

Introduction, mechanisms and machines, kinematics and kinetics, types of links, kinematic pairs and their classification, types of constraint, degrees of freedom of planar mechanism, Grubler's equation, mechanisms, inversion of four bar chain, slider crank chain and double slider crank chain.

Velocity analysis:

Introduction, velocity of point in mechanism, relative velocity method, velocities in four bar mechanism, instantaneous center.

Acceleration analysis:

Introduction, acceleration of a point on a link, acceleration diagram, Cariole's component of acceleration, crank and slotted lever mechanism.

UNIT-2

8

Cams

Introduction, classification of cams and followers, cam profiles for knife edge, roller and flat faced followers for uniform velocity, uniform acceleration,

Gears and gear trains

Introduction, classification of gears, law of gearing, tooth forms and their comparisons, systems of gear teeth, length of path of contact, contact ratio, minimum number of teeth on gear and pinion to avoid interference, simple, compound, reverted and planetary gear trains, sun and planet gear train.

UNIT-3

8

Force analysis:

Static force analysis of mechanisms, D'Alembert's Principle, dynamics of rigid link in plane motion, dynamic force analysis of planar mechanisms, piston force and crank effort. Turning

moment on crankshaft due to force on piston, Turning moment diagrams for single cylinder double acting steam engine, four stroke IC engine and multi-cylinder engines, Fluctuation of speed, Flywheel.

UNIT-4

8

Balancing:

Introduction, static balance, dynamic balance, balancing of rotating masses, two plane balancing, graphical and analytical methods, balancing of reciprocating masses,

Governors:

Introduction, types of governors, characteristics of centrifugal governors, gravity controlled and spring controlled centrifugal governors, hunting of centrifugal governors, inertia governors. Effort and Power of governor.

UNIT-5

8

Brakes and dynamometers:
Introduction, Law of friction and types of lubrication, types of brakes, effect of braking on rear and front wheels of a four wheeler, dynamometers, belt transmission dynamometer, torsion dynamometer, hydraulic dynamometer.

Books and References:

1. Kinematics and dynamics of machinery: Wilson and Sadler, Third edition, Pearson.
2. Theory of Mechanisms and Machines: AmitabhaGhosh and Ashok kumarMallik, Third Edition Affiliated East-West Press.
3. Theory of Machines and Mechanisms: Joseph Edward Shigley and John Joseph Uicker, Jr. Oxford University Press
4. Kinematics and dynamics of machinery: R L Norton, McGraw Hill
5. Theory of Machines: S.S. Rattan, McGraw Hill
6. Theory of Machines: Thomas Bevan, CBS Publishers.

RAE-601: AIRCRAFT STRUCTURE

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UNIT-110

Introduction to Aircraft Structure- Introduction to Composites. Definition and Introduction. Reinforcement: Materials and Forms. Reinforcement: Materials, Matrix Materials. TerminologiesApplications. Fabrication Processes. Fabrication Processes. Concepts of Solid Mechanics. Deformation of a body. All Stresses on the body. Basic elasticity or Hooke's law, stresses and strains, plane stress and plane strain problems.Aircraft components as fuselages, wing ribs wing spar etc. fuselage frames and. Trusses analysis of structures. The loads acting on the aircraft at different flying conditions, Guest loads Inertia forces. V-n diagram and salient features of the V-n diagram.

UNIT-29

Analysis of Fuselage or Structural idealization. Arrystress function. Fatigue loads, Fluctuating stress and theories of failure.2-D and 3-D stress and strain relations. Principal stresses. Constitutive Relations for Orthotropic Materials and Stress-Strain Transformations. Stiffness, Compliance Transformation and Hygro-thermo-elastic Constitutive Relation. Plane Stress Constitutive Equations. 2-Dimensional Lamina Analysis. Lamina Engineering Constants. Laminate Theory.Equilibrium Equations. Constitutive Equations, 3D Constitutive Equations. Constitutive Relations: Generally Anisotropy to Orthotropic. Constitutive Relations: Transverse Isotropy and Isotropy.

UNIT-38

Deflection of Beams-Types of beams. Various methods of slope-deflection for beams. Principle of virtual work.Application to deflection problems on fixed and cantilever beams.Deflection of open and close section beams. Shear force and bending moment distribution for semi-cantilever and other types of (wings and fuselage,) beam.Pure bending and Torsion Equation and their applications.Bending of thin plates, plate subjected to bending and twisting.Bending stresses in beams of unsymmetrical sections-and symmetric sections with skew loads. Combined bending and twisting on plate and beams.

UNIT-4 10

Sear Flow in Open Sections-Thin walled beams, concept of shear flow. Shear center and center of twistand its determination – shear flow distribution in symmetrical and unsymmetrical thin walled sections.Shear flow variation in idealized sectionsTorsion of closed and open section beams.Membrane analogy for circular and non-circular section.Prandtl stress function and torsion equation.

Sear Flow inClosed Sections- Bredt-Batho theory – single-cell and multi-cell tubes subject to torsion. Shearflow distribution in thin-walled single & multi-cell structures subject to combined

bending torsion with walls effective and ineffective in bending. Shear resistant web beams-Torsion field web beams.

UNIT-5 8

Bucking in Beam & Plates-Euler buckling of columns and beams inelastic stability of columns, effect of initial imperfections, energy method for the calculation of buckling loads in columns and thin plates. Flexural and tensional buckling in the thin walled columns.Buckling stress of thin walled sections, crippling strength estimation.Buckling in stiffened rectangular sheets under compression.

Books and Reference:

1. Megson T M G, 'Aircraft Structures for Engineering Students', Edward Arnold,1995.
2. Bruhn. E.H., 'Analysis and Design of Flight Vehicles Structures', Tri-state offset company, USA, 1985.
3. Howard D Curtis, 'Fundamentals of Aircraft Structural Analysis', WC BMcGraw Hill, 1997.
4. Rivello, R.M., Theory and Analysis of Flight Structures, McGraw Hill, 1993.
5. Peery, D.J., and Azar, J.J., Aircraft Structures, 2nd edition, McGraw – Hill, N.Y., 1999.
6. D.J.Perry, Aircraft Structures, Mc Graw Hill 1950
7. E.E. Sechler& L,G, Dunn, Airplane Structural Analysis & Design , Wiley & Sons Ltd,
8. F.E. Bruhn Analysis & Design of ?Flight Vehicle Structures , Tristate offset Co, 1965
9. A.S. Niles & J,S, Newel , Airplane Structural Analysis and Design, Vol & II Wiley & Sons ltd,
10. F.R. Shanley , Weight Strength Analysis of Aircraft Structures , Dover Publication , 1960
11. S. Timoshenko and J,N, Goodier , Theory of Elasticity , Mc Graw Hill , 1951
12. J.P. Den Hartog, Advanced Strength of materials, McGraw Hill , 1952
13. S. Timoshenko and J,M, Gere , Theory of Elasti Stability , Mc Graw Hill 1961
14. S. Timoshenko and K,S, Woenowsky , Theory of Plates & Shells , Mc, Graw Hill , 1959
15. M.F. Rubinstein , Matrix computer Analysis of Structures , Prentice Hall 1966
16. B. Venkatraman and S,A> Patel , Structural Mechanics with Introduction to Elasticity and Plasticity , Mc Graw Hill 1965
17. J. W Dallyand , W,F Riley Experimental Stress Analysis, Mc Graw Hill 1965
18. F.R. Shanley, Weight Strength Analysis of Aircraft Structures, Dover Publications, 1960

RAE- 602: AERODYNAMIC-I

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UNIT-1 **10**

Two Dimensional Ideal Fluid Flow: Continuity and momentum equations, Linear and angular deformations, angular velocity, Circulation and vorticity. Irrotational flows. Stream and potential functions. Varies flow as uniform flows, sources, sink, vortex flows and doublet flow. Kelvin Theorem. Kutta-Joukow-ski theorem for lift. Aerodynamic forces. Types of Airfoils and their Nomenclature.

UNIT-2 **8**

Complex Potential: Its properties and applications to various flows such as vortex doublet, corner flow, full and half body Rankin, stationery and rotating cylinder. General expressions for lift and Drag on body According to Blasius Theorem. Types of Vortex, interaction and superimpose of vortices. Induced velocity due to vortices interaction. Mapping of flow fields. Determination of pressure and velocity distributions on airfoil. Joukow-ski transformation theorem. Joukow-ski transformation of circles into Circular arcs, elliptical airfoil, Symmetrical and cambered airfoils.

UNIT-3 **8**

Thin Airfoil Theory (2-D Wing theory): Circulation and the generation of Lift and on thin airfoil, Kutta condition on thin airfoil, thin airfoil theory, Lift and pitching moment coefficient of flat and curved plates andairfoils. Effects flap on aerodynamic coefficients. Determination of mean camber line, shapes for uniform and linear distribution of circulation. Bound vortex and starting vortex, Downwash and induced drag. Helmholtz's theorems. Aerodynamic center and center of pressure.

UNIT-410

3-D Wings Theory- Vortex system around wing, horse shoe vortex. Helmholtz's theorems. Biot Savart's law, Induced velocity (Downwash) of infinite, semi-infinite and finite filaments. Prandtl's classical lifting line theory, fundamental equations, elliptic lift distribution, influence of aspect ratio on lift and drag, drag polar and lift correlation to aspect ratio. Techniques for general span wise distribution, calculation of lift and vortex induced drag, numerical problems based on above. Panel methods: General description of the panel methods.

UNIT-510

Viscous Flow Theory: Navier- Stroke's equations, Exact solutions, Poiseulle flow and Cociette flow. Velocity, tress and pressure distribution of their flows. Boundary layer theory. Blasius solution of laminar flow. Laminar and turbulent boundary layers. Boundary layer separation and control. Suction, blowing of Laminar flow Boundary layer thickness and momentum thickness of laminar and turbulent. Safe factor, average skin friction coefficients and Drag force due to skin friction coefficients.

Books and References:

1. E.L. Houghton and A.E. Brock, Aerodynamics for Engineering Students.
2. W.F. Durand (Editore), Aerodynamics Theory, Vols. I to VI, Dover Publications, 1963.
3. A.H. Shapiro, Dynamics and Thermodynamics of Compressible Flow Volts. I & II Ronald Press 1953.
4. John D. Anderson, Fundamentals of Aerodynamics.
5. A. Ferri, Elements of Aerodynamics of Supersonic Flow, MacMillan, 1949.
6. D.O. Dommasch, S.S. Sherby & T.F. Connolly, Airplane Aerodynamics, Sir,
7. A.M. Kuethe& J.D. Schetzer, Foundations of Aerodynamics, Wiley, 1961.
8. L.N. Milne Thomson, Theoretical Aerodynamics, MacMillan, 1966.

Department Elective -2

RAE-021: JET AIRCRAFT PROPULSION

L T P
3 1 0

UNIT-18

Introduction & Development of Jet Aircraft Propulsion.How the Aircraft Jet Engines make Thrust. Jet Engine Basic Performance Parameters.Turbojet, Reheat Turbojet and Multi-spool Engines.Turbofan, Turbo-prop and Turbo shaft engines and their characteristics.Ideal and Real Brayton cycles.Jet Engine Cycles for Aircraft propulsion. Cycle components and component performances. Performance and Design of Ramjet & Scramjet Engines. Future of Aircraft Propulsion .

UNIT-28

Euler turbine equation and its different forms,velocity triangles for a generalized turbo- machine degree of reaction in turbo machines, methods of representing velocity diagrams. velocity triangle for a stage radial equilibrium, free vortex, forced vortex and constant reaction designs; variation of blade height for the above types of blades, allowable temperature and pressure rise per stage; blade nomenclature, pressure loss coefficient , lift and drag, losses in blade passages, stall and surge, simple design examples.

UNIT-3 8

Thermodynamics of Compressors and their characteristics and applications.Axial Compressors: two dimensional analytical model. Cascade analysis; Loss and Blade performance estimation.Free Vortex theory; Single-Multi-stage characteristics.Elements of centrifugal compressor.Centrifugal Compressor characteristics: Surging, Choking and losses. Analysis of work done and efficiencies of compressors.

UNIT-4 8

Thermodynamics of Turbines and their characteristics and applications. Axial flow turbines; Turbine Blade 2-D (cascade) analysis.Multi-staging: Axial Turbine; Turbine Cooling Technology. Radial Turbine Aerodynamics & Thermodynamics; Losses.Engine off Design Operations. Analysis of work done and efficiencies, degree of reaction of turbines. Aircraft Engine component matching: Dimensional analysis. Engine component matching and Sizing.Installed Performance of Engine.Use of Ramjets and Pulsejets in Aircraft propulsion.

UNIT-5 8

Combustion Chamber-Types of combustion chambers: mechanism & parameters. Loss, Combustion efficiency; Combustion intensity.Practical combustion system; Stability.Intakes for Power plant: Transport/Military Aircraft. Requirement of a combustion chamber, Aerodynamic and chemical performance of combustion chambers. Balance energy equation. Effect of operating conditions of the performance of combustion chambers.Cooling of Flame tubes and

Fuel injection and control systems. Fuel injection atomization in simplex and duplex burners. All type of C-D Nozzles fixed and variable geometry, their operations characteristics at various altitudes and speeds their uses and Subsonic, Transonic, Supersonic Intake Designs.

Books and Reference -

1. Mechanics and Thermodynamics of Propulsion by Hill and Peterson,
2. Jet Propulsion for Aerospace Application by Hesse and Mumford,
3. Aircraft and Missile Propulsion by M,J, Zucrow, Vols, I and it,
4. The theory and Design of Gas Turbines and Jet Engines by E,T, Vincent,
5. Gas Turbine Theory by H, Cohen and G,F,C, Rogers,

RAE-022: COMPOSITE MSTERIALS AND STRUCTUTES

L T P
3 1 0

UNIT-1 **8**

Introduction to Composites. Definition and Introduction. Reinforcement: Materials and Forms. Reinforcement: Materials, Matrix Materials. TerminologiesApplications. Fabrication Processes. Fabrication Processes. Concepts of Solid Mechanics. Deformation of a body. All Stresses on the body.

UNIT-2 **8**

Equilibrium Equations. Constitutive Equations, 3D Constitutive Equations. Constitutive Relations: Generally Anisotropy to Orthotropic. Constitutive Relations: Transverse Isotropy and Isotropy. Constitutive Relations for Orthotropic Materials and Stress-Strain Transformations. Stiffness, Compliance Transformation and Hygro-thermo-elastic Constitutive Relation. Plane Stress Constitutive Equations. 2-Dimensional Lamina Analysis. Lamina Engineering Constants. Laminate Theory.

UNIT-3 **8**

Introduction to Classical Plate Theory. Laminate Constitutive Relations. Laminate Engineering Constants. Hygro -thermal Laminate Theory. Failure and Damage. Damage Mechanisms in Unidirectional Composites. Macroscopic Failure Theories.

UNIT-4 **8**

Tensile Fiber Mode Compressive Fiber Mode. Strength of Materials Approach. Concepts of Equivalent Homogeneity, Volumetric Averaging and Standard Mechanics. Hill's Concentration Factors Approach. Homogenization. Background of Concentric Cylinder Assemblage Model and Axial Modulus and Shear Testing. Shear and Flexural Testing.

UNIT-5 **8**

Effective Plane Strain Bulk Modulus. Effective Axial Shear Modulus. Effective Transverse Shear Modulus. Tanaka and Halpin -Tsai Models. Background to Mechanical Testing. Quality Assessment and Physical Properties. Tensile and Compressive Testing. Design Considerations in Laminated Composites. Design Considerations in Laminated Composites. Poisson's Ratio.

Books and Reference:

1. Rivello, R.M., Theory and Analysis of Flight Structures, McGraw Hill, 1993.
2. Peery, D.J., and Azar, J.J., Aircraft Structures, 2nd edition, McGraw – Hill, N.Y., 1999.
3. D.J.Perry, Aircraft Structures, Mc Graw Hill 1950
4. E.E. Sechler& L,G, Dunn, Airplane Structural Analysis & Design , Wiley & Sons Ltd,

5. F.E. Bruhn Analysis & Design of ?Flight Vehicle Structures , Tristate offset Co, 1965
6. A.S. Niles & J,S, Newel , Airplane Structural Analysis and Design, Vol & II Wiley & Sons ltd,
7. F.R. Shanley , Weight Strength Analysis of Aircraft Structures , Dover Publication, 1960
8. S. Timoshenko and J,N, Goodier , Theory of Elasticity , Mc Graw Hill , 1951
9. J.P. Den Hartog, Advanced Strength of materials, McGraw Hill , 1952
10. S. Timoshenko and J,M, Gere , Theory of Elasti Stability , Mc Graw Hill 1961
11. S. Timoshenko and K,S, Woenowsky , Theory of Plates & Shells , Mc, Graw Hill , 1959
12. M.F. Rubinstein , Matrix computer Analysis of Structures , Prentice Hall 1966
13. B. Venkatraman and S, A. Patel , Structural Mechanics with Introduction to Elasticity and Plasticity , Mc Graw Hill 1965.

RAE -023: AIRCRAFT MATERIALS &NON DESTRUCTIVE TESTING

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UNIT-1 **8**

Board classification of aircraft materials: Ferrous materials, nonferrous materials and alloys, ceramic materials and fiber reinforced composite materials, polymers, metal matrix particulate.Engineering Materials, Structural properties of materials, Atomic and lattice structure, bonding in Solids, Imperfections in crystals, Solid phase and phase diagrams.

Furnishing Materials: Plastic, wood, plywood, glue, dopes and rubber used in aircraft manufacture. Paints, surface finishes and materials.

UNIT-2 **8**

Mechanical properties and testing: Isotropy, Orthotropic, True stress and strain, Strength and elasticity, Stiffness, Resistance, Plasticity, Ductility, Toughness and Hardness of materials. Concept of Fatigue and Creep.Mechanical Testing.Factors Affecting Strength.Deformation, Plasticity and Viscouselasticity, Fracture. Heat treatment, Chemical, thermal and Technological Properties.Methods of testing and storage.

UNIT-3 **8**

Specifications: Indian Standard, British, American, French, German, and International specifications. Corrosion of material, its detection and prevention. Protective finishes. Testing Destructive and non- destructive testing techniques. Crack detection, inspection of parts by hot oil and chalk, dye-penetrate, fluorescent and magnetic particles, X-ray, ultrasonic, eddy current andacoustic emission methods.

UNIT-4 **8**

Importance of NDT in quality assurance. Different types of non-destructive techniques to obtain information regarding size, location and orientation of damage or cracks. Visual inspection techniques coin tapping technique for composite structures and adhesive bonds. Ultrasonic testing: Pulse echo technique, pitch-catch technique, through transmission technique, A-scan B-Scan the methods of NDT and highlight its role in quality assurance. The emphasis should also be on its application during the process of design, manufacturing and maintenance.

UNIT-5 **8**

X-ray radiography: Absorption spectra, short wave length, X-ray for detection of voids. Die penetration technique. Magnetic particle testing.

In each of the above techniques-

(i)Theory and basic principles, (ii) advantages/disadvantages, (iii) material of parts that can be inspected, (iv) Physical size and shape limitation, (v) economics of process, (vi) types of defects that can be detected, (vii) environment limitation are to be discussed along with equipment used for each of the techniques.

Books and References:

1. S K HajraChowdhary, Materials, Science and Engineering Processes, Media Promoters
2. George E. F. Titterton, Aircraft Materials, English Book Stores, Delhi
3. M L Begman, Manufacturing Processes, Asia Publishing House, Bombay.
4. Nondestructive Testing, Edword Arnold U.K.
5. Introduction of Nondestructive testing - A training guide, John Wiley & Sons.
6. Donglas C Lalia, NDT for Aircraft, Jeppesen.
7. NDT and Ultrasonic Testing for Aircraft, FAA-AC 43-3

RAE -024: AIRCRAFT EVALUATION

LTP
310

UNIT–1 8

Aerodynamics & performance: International Standard Atmosphere and its significance pressure and density altitudes. Compressibility effect, Mach number and its variation with altitude and temperature Take-off and landing, single engine performance at altitude, climb and descent performance at altitude, Effect of horizontal speed during climb and decent on performance and economics , Cruise performance cruise at constant speed (or Mach No.) Constant altitude, constant angle of attack and their effect on block speed and economics Noise criteria regulatory requirements regarding noise inside the passenger cabin and at near aircraft.

UNIT-2 8

Aircraft Equipment: Cockpit layout and instrumentation, automatic landing system, Air Data Computer, ICAO Landing Categories, Communication and navigation equipment and their functions in general. Flight Recorder and Cockpit Voice Recorder and related regulatory requirements and there passenger amenities like passenger address system, crew intercom; music reproducer etc., Emergency facilities equipment and related regulation.Weather radar and its uses.Environmental control, air conditioning and pressurization their significance and necessity.

UNIT-3 8

AircraftScheduling: Factor affecting airlines schedules, Commercial operation, Technical Metro logical, Airport Facilitation- run –way Strength and related requirements, Load Classification. Criteria for runaway and aircraft, Air Traffic Control and other number and other ground Communication /navigation facilities-their relationship and effect on related aircraft equipment Airport emergency Facilities – Firefighting, First Aid etc. Total take-of weight Balance diagram, calculation of c.g. and it's variation due to variation pay load and consumables.

UNIT-48 **Performance Estimation:** Power / Thrust availed and required Effect of altitude and forward speed on engine performance and power / thrust required, Level Flight performance, maximum minimum and optimum speeds, maximum range and endurance, Maximum rate of climb and its variation with altitude, Absolute and service ceilings, takeoff and landing distances, Effect of ambient temperature and wind on landing and takeoff distances, One engine take off for multi-engine civil air planes.

Longitudinal Stability- Preliminary calculations of horizontal tail setting, Static stability and static margin, V-N Diagram: Gust and maneuverability envelope.

UNIT-5 8

Principles of Aerodynamic Testing: Brief history of the development of different types of Aerodynamic testing facilities, Drop tests, Rotating f tests. low speeds high sub- sonic speed transonic supersonic and hyper – sonic wind tunnels. Wind tunnel types according to fabrications. Description and principle of operation, types of mode tests possible areas of application and limitations of each of the facilities. Basic principles of flight testing and performance reduction.

Books and Reference:

1. G. Corning , Supersonic and Subsonic Airplane Design, Boz No, 14 , College park , Maryland, 196,
2. K.D Wood Aerospace Vehicle Design Vol, –I Aircraft design, Johnson Publishing Co.
3. A. Lebedinski, Lecture notes on Aircraft Design Department of Aeronautical Engineering, Indian Institute of Science, Bangalore
4. F.K. Teichman, Air plane Design Manual , Sir Issac Pitman & sons Ltd, 1950
5. C.D. Perkins and R,E, Hage, Airplane Performance Stability and Control, John Wily and Sons inc, 1963
6. D.O. Dommasch, S,S, Sherby& T,F, Connolly, Airplane Aerodynamics , Sir Issac Pitman & sons Ltd,1961
7. I.H. Abott& A,F, Von Doenhoff, Theory of Wing Section, Dover Publication Inc, 1959,

RME----: THEORY OF MACHINES LAB

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0 0 2

Minimum 8 out of following such Experiments (or such Experiments)

1. Study of simple linkage models/mechanisms
2. Study of inversions of four bar linkage
3. Study of inversions of single/double slider crank mechanisms
4. Experiment on Gears tooth profile, interference etc.
5. Experiment on Gear trains
6. Experiment on longitudinal vibration
7. Experiment on transverse vibration
8. Experiments on dead weight type governor
9. Experiment on spring controlled governor
10. Experiment on critical speed of shaft
11. Experiment on gyroscope
12. Experiment on static/dynamic balancing.
13. Experiments on Break
14. Experiments on clutch.

RAE-651: AIRCRAFT STRUCTURE LAB

L T P
0 0 2

Minimum 10 out of following such Experiments (or such Experiments)

1. Determination of young's modulus for metallic materials
2. Determination of flexural strength of metallic materials.
3. Determination of deflection of a simply-supported beam
4. Determination of deflection of a cantilever beam.
5. Verification of superposition theorem
6. Buckling load estimation of slender eccentric columns
7. Determination of flexural rigidity of composite beams
8. Determination of Shear failure of bolted and riveted Joints
9. Determination of Truss and frame analysis.
10. Determination of symmetrical Bending of a Cantilever Beam.
11. Determination of combined bending and Torsion of a Hollow Circular Tube.
12. Determination of shear Centre of a Channel Section and I section.
13. Fabrication of a Composite Laminate.
14. Determination of Tension field beam.
15. Study of non-destructive testing procedures.
16. Free Vibration of a Cantilever Beam.
17. Forced Vibration of a cantilever Beam.

RAE-652: AIRCRAFT PROPULSION LAB

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Minimum 10 out of following such Experiments (or such Experiments)

1. To demonstrate the constructional arrangement and operation of turbojet and turbo-fan.
2. To demonstrate the constructional arrangement and operation of turboprop and turbo-shaft.
3. Recognition of visual defects of jet engines.
4. To demonstrate the operating principal 2-stroke and 4-strock CI engines.
5. To demonstrate the operating principal 2-stroke and 4-strock SI engines.
6. To demonstrate the operatingprincipal of air cooled and water cooled piston engines.
7. To demonstrate the engine configuration and firing order.
8. To demonstrate the working of engine starting systems.
9. To demonstrate the installation and removal procedure of propellers.
10. To demonstrate operation of fixed and variable pitch propellers.
11. To demonstrate operation of turbine blades to turbine disc.
12. To demonstrate the working of exhaust gas temperature measurement system.
13. Calculate the cylinder area, sweep area and compression ratio for C I engine.
14. Test the on piston engine determine: Specific fuel consumption, Break horse power, Indicated horse power for four stroke petrol engine.
15. Test the on piston engine determine: Break thermal efficiency, Mechanical efficiency. Volumetric efficiency for four stroke petrol engine.
16. To determine Velocity and pressure measurements in co-axial jets.
17. To steady of Wall pressure distribution in subsonic diffusers.

To steady of Flame stabilization studies using conical flame holders

RAE-653: DESIGN OF AIRCRAFT ELEMENTS& SIMULATION LAB

L T P

0 0 2

Minimum 10 out of following such Experiments (or such Experiments)

1. Design of Aircraft components subjected to combined steady and variable loads
2. Design of eccentrically loaded riveted joint.
3. Design and drafting of riveted joints
4. Design and drafting of lap joint and butt joints
5. Design of shaft for combined constant twisting and bending loads
6. Design and drafting of welded joints.
7. Design and drafting control components gear.
8. Understand layout of typical wing structure.
9. Understand layout of typical fuselage structure.
10. Understand layout of typical control system
11. Understand computer added modeling of layout of typical wing structure
12. Understand computer added modeling of layout of typical fuselage structure.
13. Understand computer added modeling of layout of typical control system
14. Understand computer added modeling of layout of typical landing gear.
15. Three view diagram of a typical aircraft